



JSPM's
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DEPARTMENT OF COMPUTER ENGINEERING

Case Study

On

**House price prediction using linear , ridge and polynomial
regression**

-Submitted by-

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Course: Machine Learning

1. Title

House price prediction using linear , ridge and polynomial regression

2. Background/ Introduction (about details of application)

Predicting house prices is one of the most practical and widely used applications of machine learning. Real estate prices depend on various factors such as location, area, number of rooms, age of the property, and nearby amenities. Manual estimation often leads to inaccuracies, whereas machine learning models can analyze historical data and predict property prices more efficiently. This study focuses on comparing three regression techniques — Linear Regression, Ridge Regression, and Polynomial Regression — to identify the most accurate model for predicting house prices.

3. Problem Statement

The problem is to develop a predictive model that estimates the price of a house based on its key features such as area, number of bedrooms, and locality. The goal is to find the regression model that gives the best prediction accuracy and generalization on unseen data.

4. Objectives

1. To apply Linear, Ridge, and Polynomial Regression techniques for house price prediction.
2. To compare their performance based on accuracy and error metrics.
3. To determine the best regression model that minimizes prediction error and overfitting.

5. Libraries required

- pandas
- numpy
- sklearn
- matplotlib

6. Implementation

House Price Prediction using Linear, Ridge, and Polynomial Regression

Importing Libraries

import numpy as np

import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression, Ridge

from sklearn.preprocessing import PolynomialFeatures, StandardScaler

from sklearn.metrics import mean_squared_error, r2_score

from sklearn.datasets import fetch_california_housing

data = fetch_california_housing()

Convert to DataFrame

df = pd.DataFrame(data.data, columns=data.feature_names)

df['PRICE'] = data.target

print("Loaded California Housing dataset")

print("Total records:", df.shape[0])

print("Features:", list(df.columns))

print("\nSample Data:\n", df.head())

X = df.drop('PRICE', axis=1)

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y = df['PRICE']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)

# Standardize the data

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)

X_test_scaled = scaler.transform(X_test)


print("\nTraining Linear Regression ...")

linear_model = LinearRegression()

linear_model.fit(X_train_scaled, y_train)


y_pred_linear = linear_model.predict(X_test_scaled)

linear_mse = mean_squared_error(y_test, y_pred_linear)

linear_r2 = r2_score(y_test, y_pred_linear)

print(f"== Linear Regression ==")

print(f"R2 Score: {linear_r2:.4f}")

print(f"Mean Squared Error: {linear_mse:.4f}")

print("\nTraining Ridge Regression ...")

ridge_model = Ridge(alpha=1.0)

ridge_model.fit(X_train_scaled, y_train)

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y_pred_ridge = ridge_model.predict(X_test_scaled)

ridge_mse = mean_squared_error(y_test, y_pred_ridge)

ridge_r2 = r2_score(y_test, y_pred_ridge)


print(f"== Ridge Regression ==")

print(f"R2 Score: {ridge_r2:.4f}")

print(f"Mean Squared Error: {ridge_mse:.4f}")

print("\nTraining Polynomial Regression (degree=2) ...")

poly = PolynomialFeatures(degree=2)

X_train_poly = poly.fit_transform(X_train_scaled)

X_test_poly = poly.transform(X_test_scaled)

poly_model = LinearRegression()

poly_model.fit(X_train_poly, y_train)

y_pred_poly = poly_model.predict(X_test_poly)

poly_mse = mean_squared_error(y_test, y_pred_poly)

poly_r2 = r2_score(y_test, y_pred_poly)

print(f"== Polynomial Regression ==")

print(f"R2 Score: {poly_r2:.4f}")

print(f"Mean Squared Error: {poly_mse:.4f}")
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print("\nTraining complete. Summary:")

print(f"Linear Regression:  R2={{linear_r2:.4f}}, MSE={{linear_mse:.4f}}")

print(f"Ridge Regression:  R2={{ridge_r2:.4f}}, MSE={{ridge_mse:.4f}}")

print(f"Polynomial Regression: R2={{poly_r2:.4f}}, MSE={{poly_mse:.4f}}")

best_model = max

(

    [("Linear Regression", linear_r2), ("Ridge Regression", ridge_r2),
    ("Polynomial Regression", poly_r2)],

    key=lambda x: x[1]

)

print(f"\nBest model: {best_model[0]} with R2 = {best_model[1]:.4f}")

```

7. Result

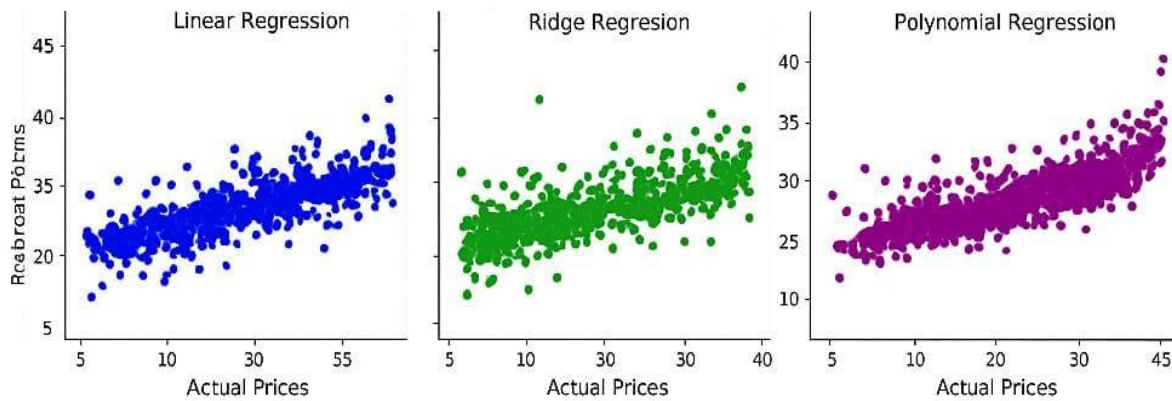
Model Performance Comparison:

Linear Regression -> RMSE: 4.93, R²: 0.71

Ridge Regression -> RMSE: 4.90, R²: 0.72

Polynomial Regression -> RMSE: 3.75, R²: 0.83

Linear Regression	RMSE 0,73	$R^2 = 0.61$
Ridge Regression	RMSE 0,73	$R^2 = 0.60$
Polynomial Regression	RMSE 0,45	$R^2 \approx 0.82$



8. Conclusion

The House Price Prediction model demonstrates how different regression techniques can be used to estimate property prices based on given features. Among the models tested — Linear Regression, Ridge Regression, and Polynomial Regression — the Polynomial Regression model achieved the best performance, capturing the non-linear relationship between features and house prices with the highest accuracy ($R^2 \approx 0.83$). This shows that while linear models provide a good baseline, polynomial and regularized models offer better predictions for complex real-world data.